

Devin Pham

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United States Citizen

SUMMARY

Robotics Software Engineer with experience building AGV software, ROS2 applications, perception pipelines, simulation workflows, and GPU deployment infrastructure. Strong in C++, Python, ROS2, PCL, YOLOv11, Docker, CUDA, Gazebo, RViz2, and real-world robotic system validation. Seeking robotics, autonomy, perception, or simulation engineering roles.

WORK EXPERIENCE

Essential Aero Inc. · Robotics Engineer

San Diego, CA | Jun 2024 – Apr 2026

- Optimized LiDAR and radar point cloud processing workflows for autonomous ground vehicles using **C++** and **PCL**.
- Developed a **YOLOv11-based human-tracking pipeline** to support AGV relocation and human-aware robot behavior.
- Built and optimized **CUDA-enabled Docker containers** for deploying GPU-accelerated perception models in production-like environments.
- Simulated, validated, and debugged robotic software workflows using **ROS2**, **Gazebo Sim**, **RViz2**, and rosbag data.
- Optimized a multithreaded ROS2 data parser, achieving an **800% improvement in processing efficiency**.
- Developed browser-based internal tools for AGV configuration and control using **HTML**, **JavaScript**, and **CSS**.
- Performed system validation, issue triage, and defect reporting using **Jira** and **Epsilon3**.

Genasys Inc. · Software Engineer Intern

San Diego, CA | Jun 2023 – Sep 2023

- Developed automation workflow for creating bootable USB drives, reducing manual setup time and improving engineering workflow efficiency by **50%**.
- Identified, reproduced, and documented software bugs for internal engineering teams using Jira.
- Assisted hardware validation efforts for custom test cables using certified diagnostic tools.
- Standardized internal testing procedures through clear technical documentation and repeatable instructions.

Genasys Inc. · Test Engineer Intern

San Diego, CA | Jun 2022 – Sep 2022

- Developed a GUI-based Python application using **Tkinter**, **NumPy**, and **pandas** to generate test parameters from internal company data, improving test accuracy by **10%**.
- Built a **C# Windows Forms application** to control an Audio Power Board via USB.
- Contributed code changes through Gitflow and Bitbucket workflows.
- Coordinated with distributed engineering teams through Slack-based technical collaboration.

EDUCATION

University of California, San Diego

San Diego, CA

B.S. Data Science

June 2024

- Focus on machine learning and artificial intelligence.
- Senior project on “Deep Learning in Autonomous Vehicles.”

SKILLS

Languages: C++, Python, JavaScript, SQL, C#

Robotics: ROS2, Gazebo Sim, RViz2, rosbag, TF, AGV Systems, Robot Telemetry

Perception: YOLOv11, OpenCV, PCL, LiDAR Point Clouds, Radar Data Processing

Deployment/Tools: Docker, CUDA, Linux, Git, Jira, Epsilon3

Web/Systems Tools: HTML, CSS, Next.js, FastAPI, Express

Data: pandas, NumPy, PostgreSQL, MySQL

PROJECTS

Camera-Based Human-Following AGV System | *Python, ROS2, TF, YOLOv11, Gazebo, RViz2* [\[link\]](#)

2025

- Developed and deployed a ROS2-based perception pipeline enabling a physical AGV to detect, track, and follow a target human in real-world testing.
- Integrated YOLOv11 detection outputs with robot control logic, using RViz2 and rosbag data to validate tracking behavior and debug system performance.

LiDAR-Based Planar Surface Detection for Robotic Process Monitoring | *C++, ROS2, PCL, RANSAC, LiDAR* [\[link\]](#)

2025

- Developed a ROS2-based LiDAR perception pipeline to detect FOD mat misalignment for an autonomous airport debris-removal robot during disposal-station entry.
- Implemented ROI preprocessing, Crop Hull filtering, and RANSAC plane segmentation to estimate left and right mat orientations and trigger stop-or-proceed control logic based on angle deviation.